

Hospital Emergency Triage System

CMPS 445 Project

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Introduction

Hospital handle many patients with different levels of urgency.

Triage systems help decide who should be treated first.

Our project simulates a simple emergency triage workflow.

Written in Python.

Focus on problem-solving, menu dedign, logic, and data handling.

Project Goals

1

Build a functional triage system using basic Python concepts.

2

Let staff register and organize patients.

3

Create a clean, beginner-friendly application.

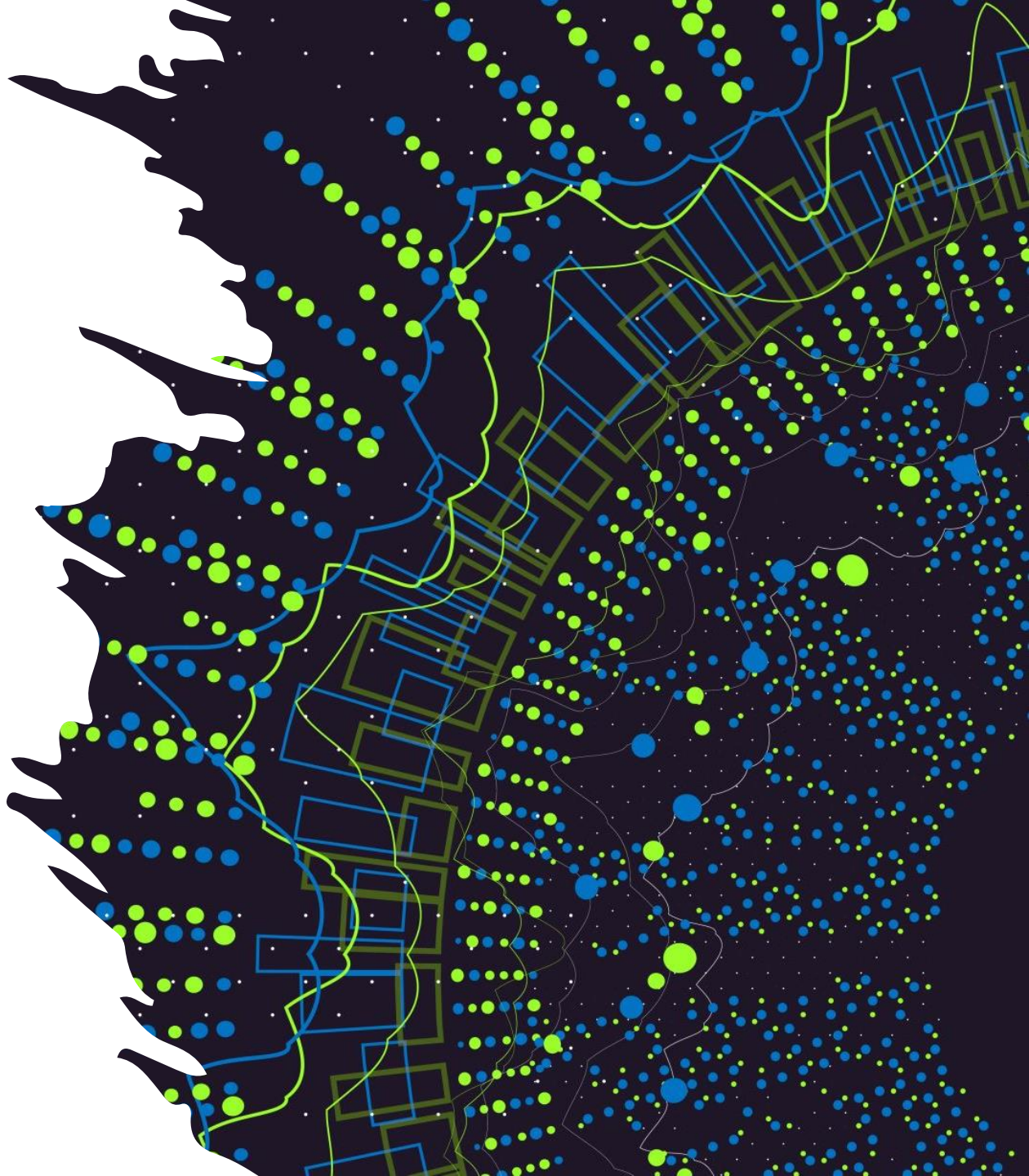
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Demonstrate real life decision making based on symptoms.

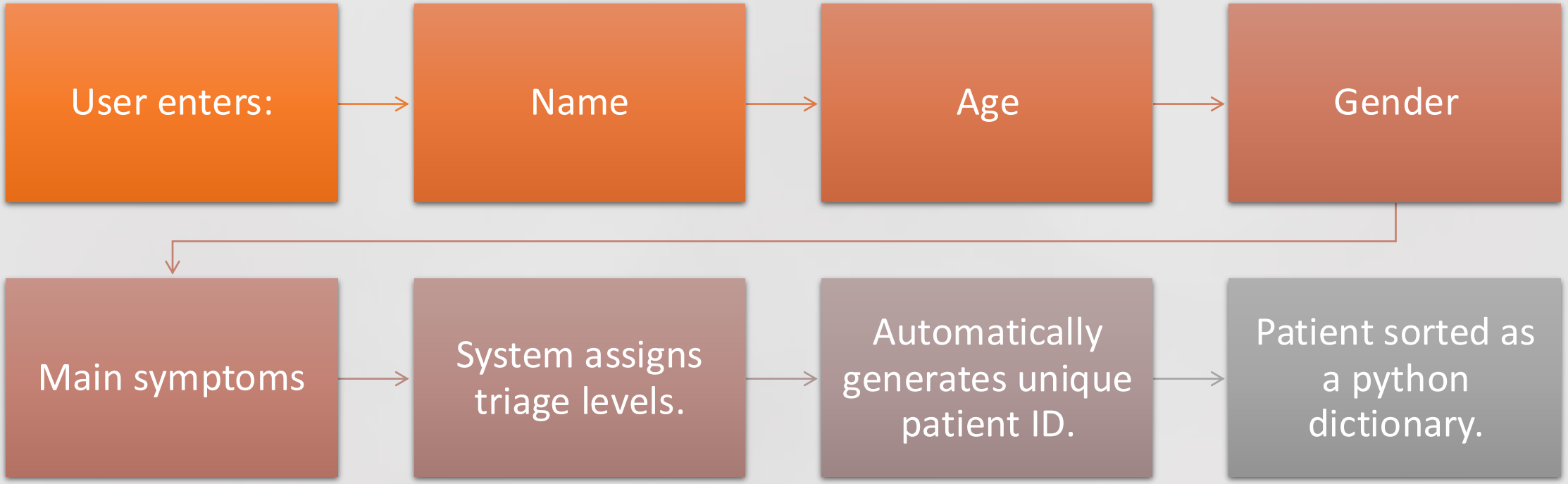
Features Overview

Our system supports:

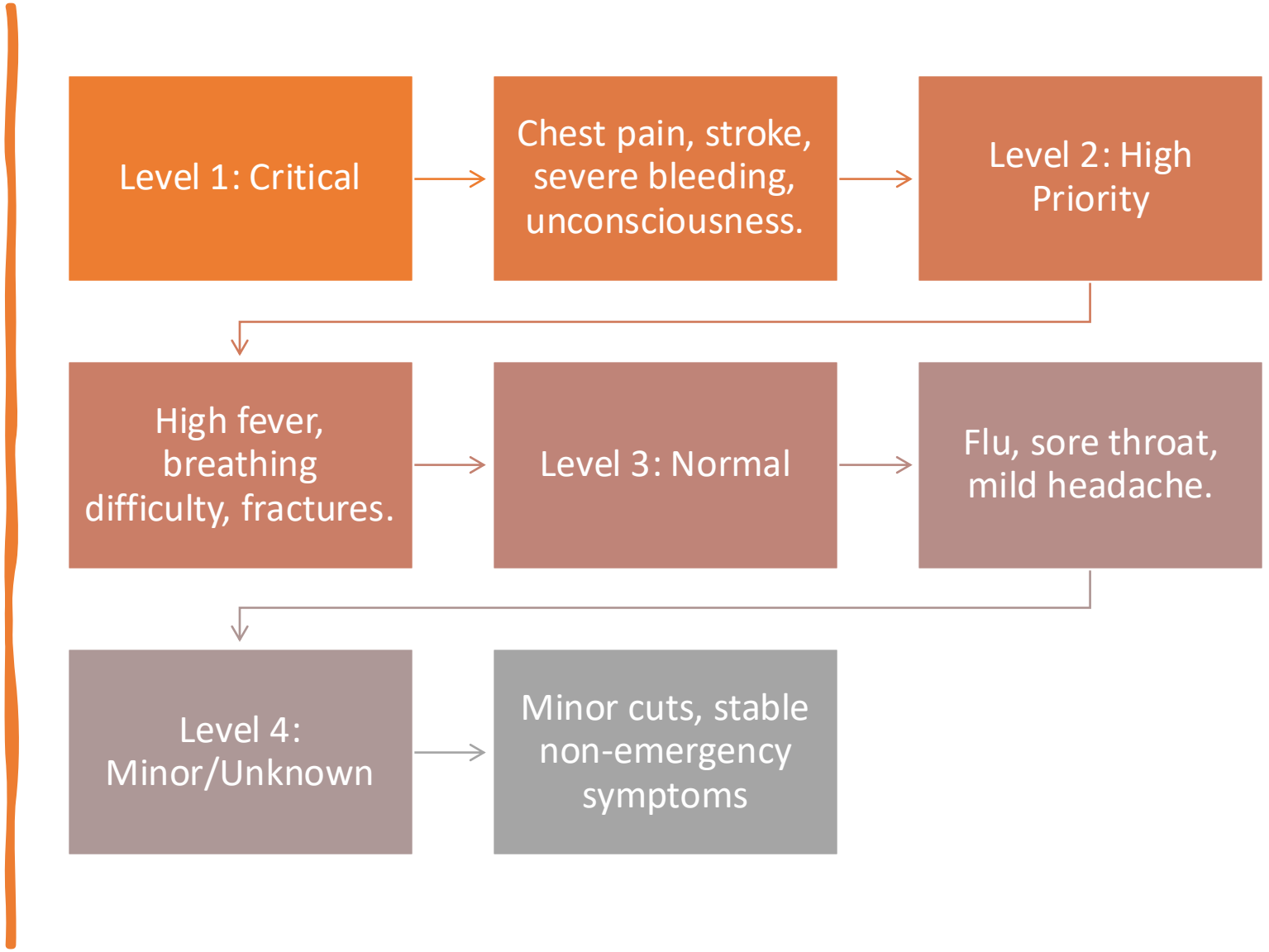
- Add new patients.
- Auto assign triage levels (1 to 4).
- View all patients sorted by severity.
- Search by name or symptoms.
- Edit patient information.
- Discharge patients.
- View statistics.
- Triage help menu



Adding a Patient



Triage Level



Data Structure

This allows:

- Easy searching
- Editing
- Sorting
- Displaying in menu

Sorting Logic

Patients are displayed by:

- Triage level
- Age



Searching & Editing

Searching:

- Keyword based search by name or symptom.
- Case insensitive

Edit:

- Use patient ID
- Change name, age, gender, symptom
- Triage level recalculated automatically



Discharging Patients

Staff enters patient ID

Patient removed from list

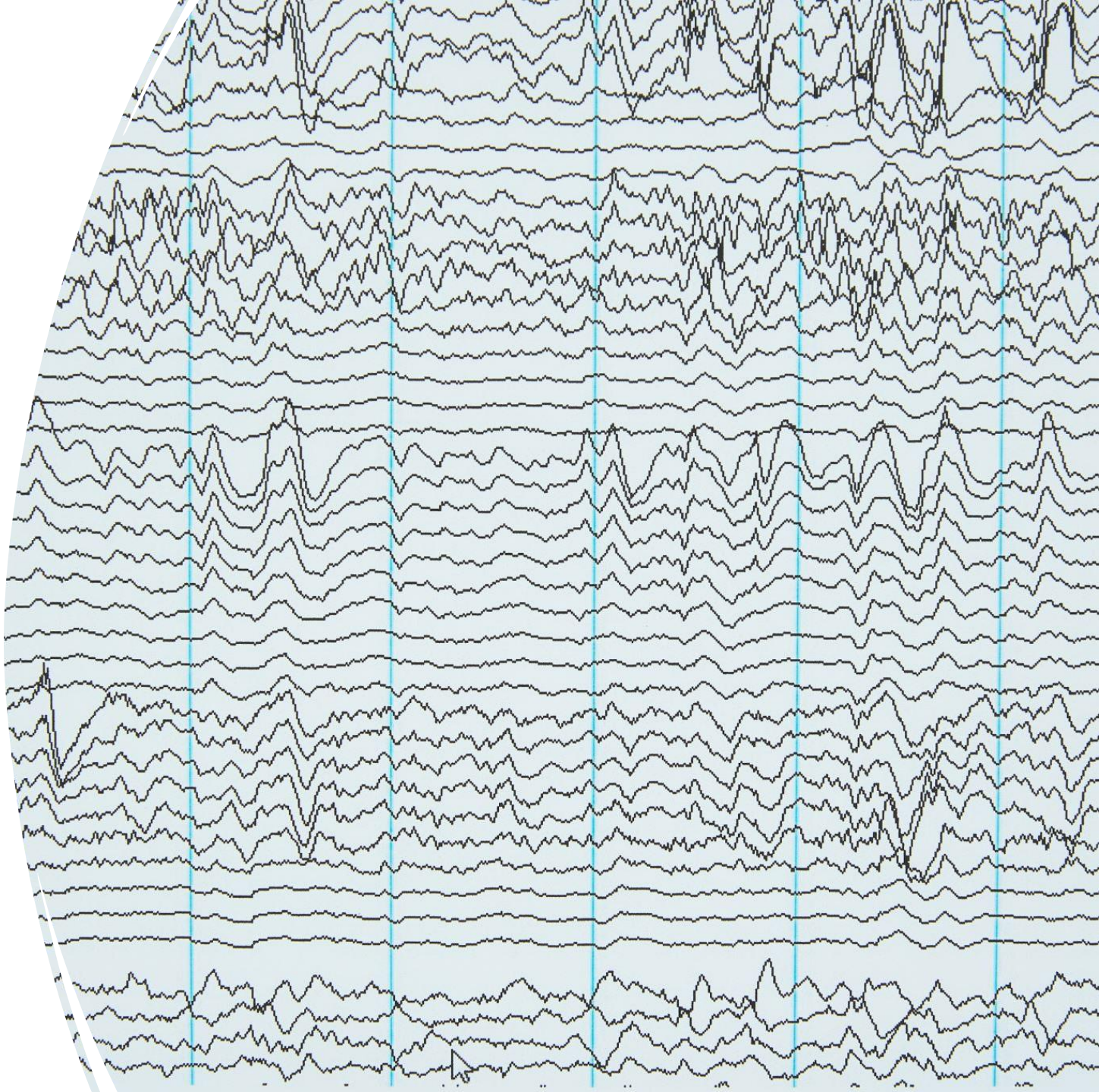
Simulates real hospital workflow



Statistics Feature

Shows:

- Total number of patients
- Number per triage level
- Average age
- Helps understand ER load and patient distribution





Language Choice: Python vs Java

We chose Python because:

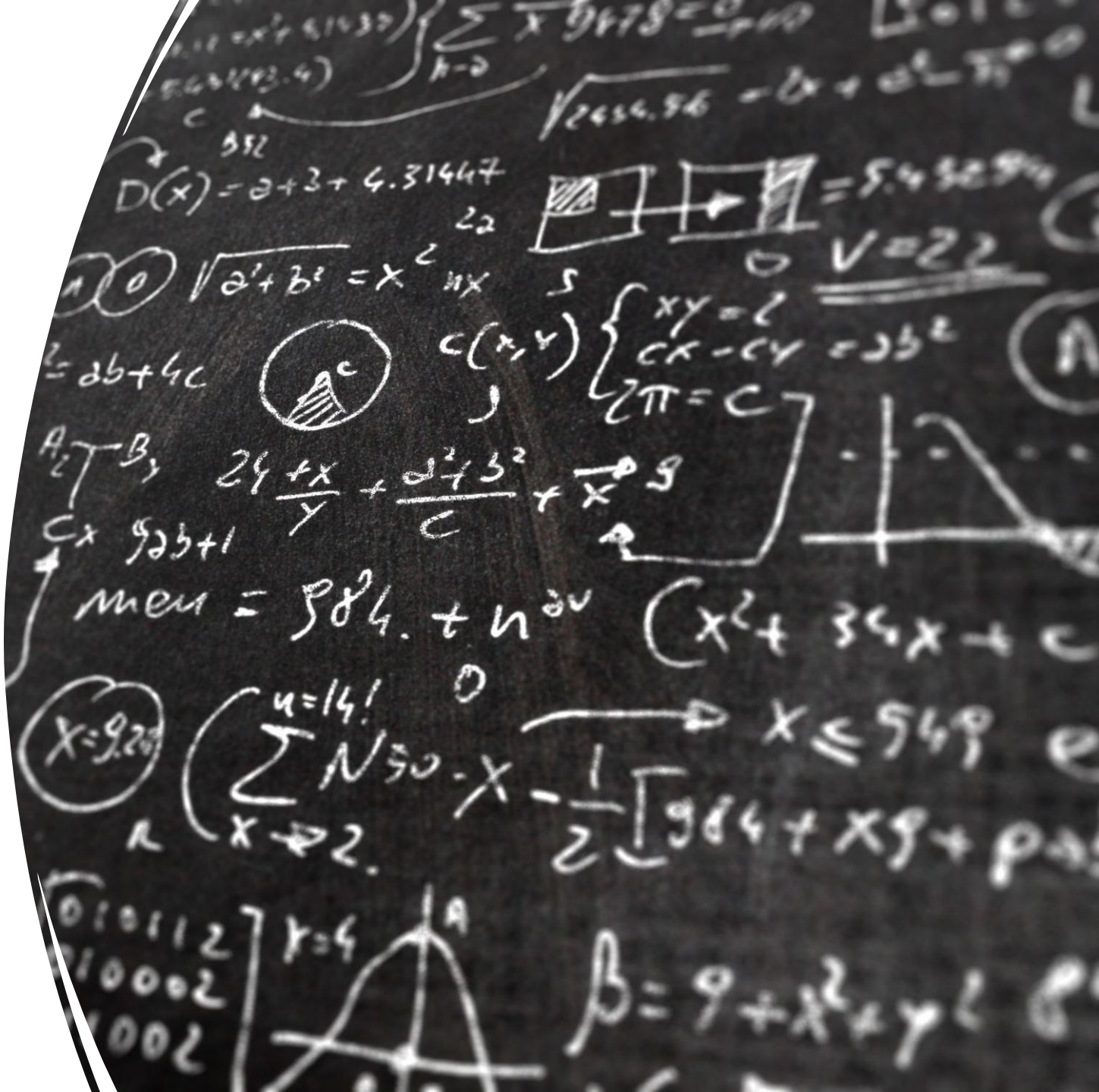
- Easier syntax for beginners
- Faster development
- No need to declare variable types
- Cleaner code for text-based apps

Java would require:

- Defining classes
- More boilerplate
- Explicit typing
- More code for simple input/ouput

Programming Concepts Used

- Variables & basic data types
- Lists
- Dictionaries
- Functions
- Loops (for, while)
- Conditionals (if/elif/else)
- Sorting
- Input/Output



Challenges We Faced

- Keeping the program simple but functional
- Designing a clear menu system
- Understanding sorting with multiple conditions
- Input validation (age, empty names, etc.)
- Organizing tasks between group members



Conclusion



We built a working Hospital Triage System using Python



The project helped us understand real world problem solving



We practiced major programming concepts from the course



Python was an excellent choice for a beginner team project



The system is easy to extend and improve

Thank you!